

Crossover Disambiguation in Kuramoto-Coupled Dense Associative Sequential Memory

Learning rules for resolving branch ambiguity in phase-oscillator sequence retrieval

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Outline

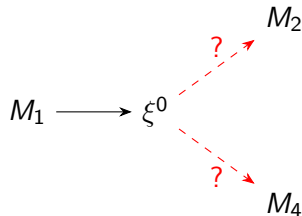
- 1 Introduction and Motivation
- 2 Literature Review
- 3 Model and Learning Rules
- 4 Numerical Experiments and Results
- 5 Theoretical Analysis
- 6 Conclusions and Outlook
- 7 Active Questions for Further Investigation

Sequential associative memory

- Hopfield memories store *static* attractors. Motor sequences, speech and episodic replay require *ordered* ones.
- Sequential retrieval = a chain of transitions $\mu \rightarrow \mu + 1$.
- **[TODO: one line: hippocampal replay + phase-native hardware]**

The crossover problem

- Two sequences share a contiguous run of patterns.
- At the shared state the network configuration is *identical* either way: the history is not encoded in the state.
- Markov-1 drives both continuations equally, so noise alone picks the branch.



Can the coupling alone encode the route into ξ^0 ?

Objectives

- ① A Kuramoto phase-oscillator DAM for *sequences*.
- ② A learning rule that resolves crossover from the coupling alone — no external gating signal.
- ③ *When* is disambiguation possible at all, and how robust is it to corruption, network size and disorder?
- ④ **[TODO: one line: analytical / tensor contribution]**

From Hebbian storage to dense associative memory

Hopfield (1982) — pairwise Hebbian

$$J_{ij} = \frac{1}{N} \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}:$$

$$E = -\frac{N}{2} \sum_{\mu} (m^{\mu})^2, \quad m^{\mu} = \frac{1}{N} \sum_j \xi_j^{\mu} s_j$$

Crosstalk caps capacity at $\approx 0.14N$.

Krotov–Hopfield DAM (2016) — sharpen the nonlinearity, i.e. higher-order interactions:

$$E = - \sum_{\mu} F(m^{\mu}), \quad F(x) = x^{d+1}$$

Capacity $\mathcal{O}(N^d)$; minima sharpen with d .

Resulting drive

Descending E gives the *pure-power* drive $\sum_{\mu} \xi_i^{\mu} (m^{\mu})^d$ — one pattern index per term, exponent d as in Krotov DAM. Every rule that follows changes only *which* pattern each factor reads.

From the DAM energy to the Kuramoto sequence drive

- ① **Spins $\rightarrow$ phases:** $s_i = \cos \theta_i$, so $\dot{\theta}_i = -\partial_{\theta_i} E = \omega_i - \sin \theta_i \sum_{\mu} \xi_i^{\mu} (m^{\mu})^d$.
- ② **Symmetric ($\mu \rightarrow \mu$):** true gradient flow; stored patterns are minima.
- ③ **Asymmetric ($\mu \rightarrow \mu + 1$):** drive the *successor* (Sompolinsky–Kanter): stored patterns cease to be resting points and the state traverses the chain.

Absence of an energy functional

Asymmetric couplings $\Rightarrow$ no potential with $\dot{\theta} = -\nabla E$. “Minima”, “saddles”, “symmetry of the landscape” below refer to a **Lyapunov function** of the flow. The absence of gradient structure is precisely what permits transitions.

- **Hopfield (1982)** — pairwise Hebbian attractor memory.
- **Krotov & Hopfield (2016)** — dense associative memory: power- d interactions, super-linear capacity.
- **Sompolinsky & Kanter (1986)** — asymmetric $\mu \rightarrow \mu + 1$ couplings turn attractors into a sequence.
- **Kuramoto (1975)** — the phase-oscillator substrate.
- **Nägerl & Berloff (2025)**, arXiv:2507.21984 — DAM in a higher-order Kuramoto network; the base model for this work.
- **[TODO: oscillator / photonic hardware refs; Markov-chain sequence refs]**

Open problem: crossover disambiguation

No learning rule for crossover disambiguation

Existing formalisms specify *how* one pattern drives the next, but not what happens when two sequences pass through the *same* state.

- A transition matrix only ever sums pure powers, $\sum_{\beta} T_{\alpha\beta}(m^{\beta})^d$.
- It cannot represent a conjunction: *drive $\xi^{\mu+1}$ only if m_{μ} **and** $m_{\mu-1}$ are simultaneously large.*
- $\Rightarrow$ gate on a *product of overlaps at different lags.*

Motivation for Kuramoto dynamics

- **Smooth retrieval.** Continuous phases resolve the traversal of a crossover *in time*, rather than as a discrete jump.
- **Physical realizability.** Rules expressed purely as oscillator coupling are natively executable on analog and photonic hardware.
- **[TODO: theta–gamma phase coding, if used in Applications]**

Objects: patterns, overlaps, couplings

Object	Shape	Content
θ	N	oscillator phases; $s_i = \cos \theta_i$
$\xi^{a,\mu}$	$K \times P \times N$	stored patterns, $\xi_i = \cos \varphi_i \in \{-1, +1\}$, $\varphi_i \in \{0, \pi\}$
$\xi^{a,\mu+1}$	$K \times P \times N$	successor tensor: row μ holds the pattern μ drives
$m_i^{a,\mu}$	$K \times P$ (per i)	leave-one-out overlaps, $\frac{1}{N-1} \sum_{j \neq i} \xi_j^{a,\mu} \cos \theta_j$
$T_{\alpha\beta}^a$	$K \times P \times P$	transition matrix; row = destination, column = source
R (rule table)	$n_{\text{terms}} \times (d+2)$	rows $[s \mid \ell_1 \dots \ell_d \mid w]$: target offset, lags, weight
J (coupling)	rank $d+1$	$J_i^{\mu_1 \dots \mu_d, \nu}$, contracted against d overlaps

- Patterns are real, so $|m|^2 = m^2$: the write-up's fourth-order bidegree-(2,1) fields are the $d=3$ rows of R .
- T sums *pure powers* $\sum_{\beta} T_{\alpha\beta} (m^{\beta})^d$; a rule table multiplies overlaps read at *different* lags, which no T can express. The shift matrix $T_{\alpha\beta} = \delta_{\alpha, \beta+1}$ and the row $[1 \mid 0, 0, 0 \mid 1]$ are the same baseline.

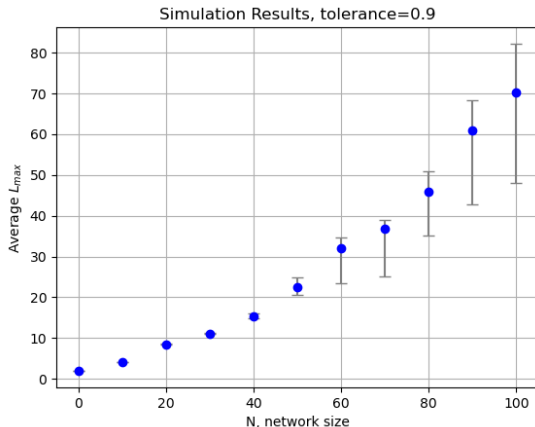
Baseline rule and retrieval algorithm

$$\dot{\theta}_i = \omega_i - \sin \theta_i \sum_{\mu} \xi_i^{\mu+1} (m_i^{\mu})^d, \quad m_i^{\mu} = \frac{1}{N-1} \sum_{j \neq i} \xi_j^{\mu} \cos \theta_j$$

Read the current overlap, drive the next pattern: Sompolinsky–Kanter at Krotov power d .

-
- 1: cue $\theta(0) \leftarrow$ (possibly corrupted) ξ^0
 - 2: **for** each step **do**
 - 3: m_i^{μ} for all μ ; $V_i = \sum_{\mu} \xi_i^{\mu+1} (m_i^{\mu})^d$
 - 4: $\theta_i += dt (\omega_i - \sin \theta_i V_i)$ (+ noise)
 - 5: **end for**
 - 6: decode the sequence from overlap peaks
-

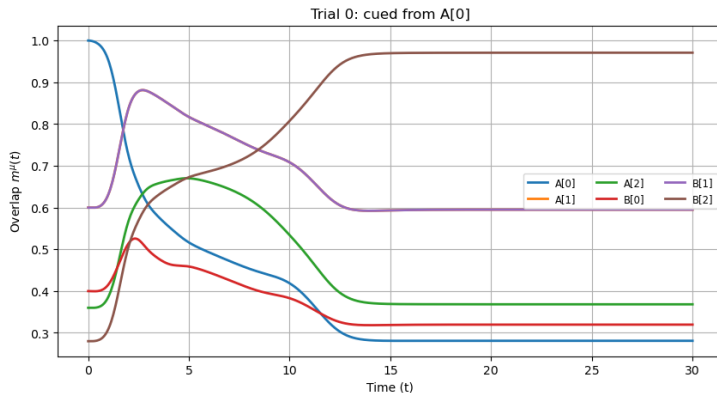
Single-sequence retrieval, baseline rule



Pattern overlaps $m^{\mu}(t)$ during single-chain retrieval under the baseline (forward) rule

[TODO: caption: clean peak-to-peak retrieval through P patterns]

Baseline rule at a crossover



Branch overlaps out of the shared state at a crossover, baseline (forward) rule

Both continuations rise together out of the shared state. Branch choice at chance: **[TODO: XX%]** vs. 50%.

Innate symmetry of the Lyapunov function

- Couplings are asymmetric $\Rightarrow$ no energy; the flow is organized by a Lyapunov function V built from the same pure powers.
 - Every term of V reads *one* pattern index — nothing in it records *how* the state reached the shared pattern.
 - At ξ^* the gate depends only on $m(\xi^*)$, identical for both branches: the two exits are related by an exact symmetry of V (a degenerate saddle).
- $\Rightarrow$ The symmetry must be broken by information the current state cannot carry: overlaps at *several lags*, multiplied rather than summed within a single term.

Heterogeneous lag-context rules

$$\dot{\theta}_i = \omega_i - \sin \theta_i \sum_{\mu} \sum_{\text{terms}} w \xi_i^{\mu+s} \prod_{l \in \text{lags}} m_i^{\mu-l}$$

- A rule is a *table of terms*: target offset s , d source lags, weight w . Lags $(0, 0, 0)$ recovers the forward rule.
- Different lags in one term $\Rightarrow$ the gate encodes *which branch* produced the current state.

How the experiments are organised

- ① **Why anything moves.** $\omega \equiv 0$ is an exact fixed point; the natural frequency is what starts the cascade, and how much of it matters.
- ② **Why branching is hard.** A shared run is bit-identical between branches; the gate must reach past it.
- ③ **Metrics, topologies, statistics.** Four crossover metrics, four geometries, 100-instance seed sweeps with spread.
- ④ **Robustness.** Network size, corruption ratio, and two knobs on the drive.
- ⑤ **Appendix.** Random patterns; settle vs. loop.

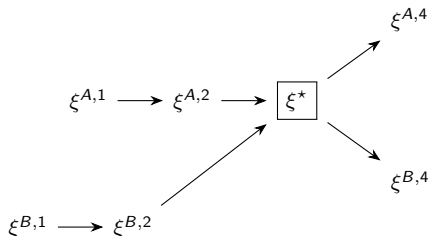
Every number below is reproduced by `results_summary.ipynb`.

Sequence construction: smooth corruption chains

- ξ^1 uniform on $\{0, \pi\}^N$; then $\xi^{k+1} = \text{corrupt}(\xi^k, \rho)$: $\lfloor \rho N \rfloor$ sites drawn without replacement, $\varphi \rightarrow \pi - \varphi$.
- Consecutive overlap is therefore fixed by construction:

$$\frac{1}{N} \sum_i \xi_i^k \xi_i^{k+1} = 1 - 2\rho, \quad d(\xi^k, \xi^{k+1}) = 2\rho.$$

- The patterns form a *chain*: neighbours stay inside each other's basin, which is what lets the cascade hop at all.
- Lagged overlaps decay geometrically, $m_{\mu-\ell} \sim (1 - 2\rho)^\ell$: hence the gain filter's $(1 - 2\rho)^{-\sum \text{lags}}$.



Shared run generated *once* and reused by both branches: those overlaps are bit-identical, so the ambiguity is exact. Each branch is an independent corruption chain off the run.

Uncorrelated patterns score 0% for every rule — nothing retrieves without the chain.

Tag	Geometry	P	Shared run L	Reach needed
2-1-2	one shared pattern	5	1	1
2-2-2	shared segment	6	2	2
2-3-2	failure case	7	3	3 (beyond catalogue)

- **Scoring:** every *unshared* pattern of the cued branch decoded, in order. Shared positions carry no rule information and are excluded.
- **Branch diagnostic:** correct vs. incorrect drive at the exit; separates incorrect selection from correct selection without lock-on.

Metrics used throughout

	metric	definition	chance / floor
Crossover — 2-1-2, 2-2-2, 2-3-2; shared positions excluded			
1	correct	every unshared pattern of the cued branch decoded <i>in order</i> , before the other branch appears	—
	crossed	the other branch appears first	—
	incomplete	the run ended before the branch resolved	—
2	overlap quality	mean peak overlap of those patterns, successful attempts only, with spread	—
3	branch-correct	did the trajectory favour the cued continuation?	0.5
4	margin	signed gap between the cued and the other continuation's peak overlap	0
Single chain — $P = 10$, no branch			
5	retrieval fraction	how much <i>of a</i> sequence came back; partial credit on a strict prefix	$1/P = 0.10$
6	strict rate	how many <i>of the</i> sequences came back whole	—

- A pattern counts as decoded at overlap $\geq$ tolerance = 0.5.
- **Metric 3 isolates the decision**; metric 1 bundles it with completion, correct $\approx$ branch-correct $\times$ (1 – incomplete).
- self is the control in both regimes, and forward the baseline at a crossover.

Catalogue of learning rules

Rule	Gate at pattern μ	Reach
self	$\xi^\mu m_\mu^3$	0 (control)
forward (baseline)	$\xi^{\mu+1} m_\mu^3$	0
skip-two	$\xi^{\mu+2} m_\mu^3$	0
bigram	$\xi^{\mu+1} m_\mu m_{\mu-1} ^2$	1
multilag	$\xi^{\mu+1} m_\mu \sum_k w_k m_{\mu-k} ^2$	$\leq K$
coh bigram	$\xi^{\mu+1} m_\mu^2 m_{\mu-1}$	1
coh trigram	$\xi^{\mu+1} m_\mu m_{\mu-1} m_{\mu-2}$	2

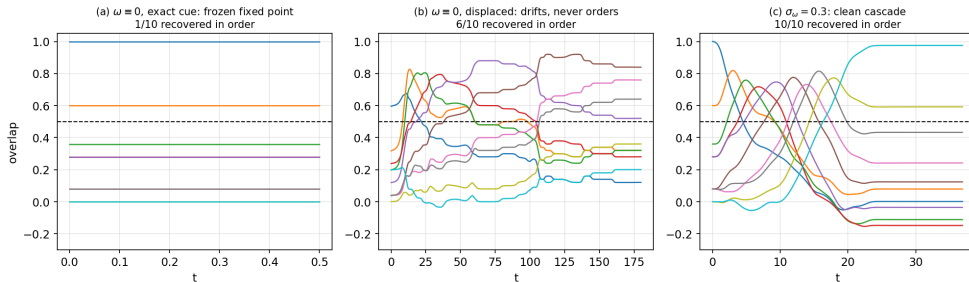
Reach $\geq L$ is the entry requirement for accessing *any* unshared information. Clearing it does not guarantee the rule can use it.

Global parameters

Symbol	Value	Symbol	Value
N	200 crossover; 100 single (swept)	dt	0.05
P	5, 6, 7 crossover; 10 single	T	80 crossover; 180 single
d	3	integrator	Euler / Euler–Maruyama
ρ	0.2 (sweep 0.05–0.35)	early stop	$\max_i \dot{\theta}_i < 10^{-6}$, 10 steps
μ_ω	0	tolerance	0.5 overlap to decode a pattern
σ_ω	0.3	decoder margin	0.01 crossover; 10^{-3} single
phase noise	0.01 (inert: ODE mode)	dwell	20 per transition
trials	20×2 cues; 100 seed sweep	seeds	pool of 10^4

- $\omega \neq 0$ is load-bearing — see next slide.
- Raw and gain-corrected catalogues are run at every operating point; the gain is rebuilt at each ρ of a sweep.
- **Two base sizes.** Branch choice saturates at $N = 200$, so the crossover experiments run there. The single-chain sweeps keep $N = 100$ as their base and sweep N explicitly — that sweep *is* the experiment.
- **Caveat:** every network uses seed=10, so one ω draw is reused throughout. On the ω axis every result is $n = 1$.

1. Why anything moves: the natural frequency

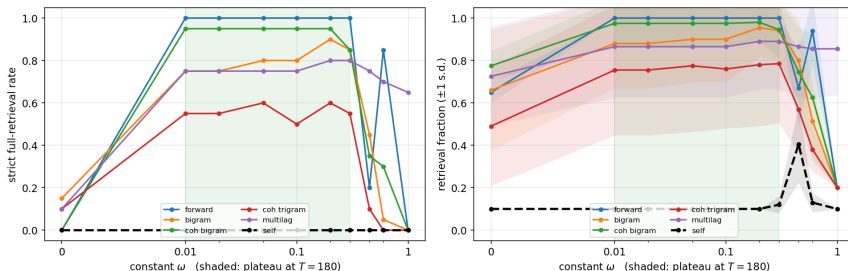


Retrieval at $\omega \equiv 0$ from an exact cue (a) and a displaced cue (b), against $\sigma_\omega = 0.3$ (c)

Note the differing time axes: (a) exits at the stationary early-stop within a few steps.

- At an exact cue every $\theta_i \in \{0, \pi\}$, so $\sin \theta_i = 0$ and the coupling $-\sin \theta_i V_i$ vanishes *identically*, however large V_i is. **(a)** sits at an exact fixed point — the drive is degenerate. The 1/10 it scores is the cue itself.
- **(b)** displaced start (flips *plus* angular jitter, off-lattice). No reliable cascade.
- **(c)** $\sigma_\omega = 0.3$: a clean cascade, each pattern peaking in turn and the full chain recovered in order.

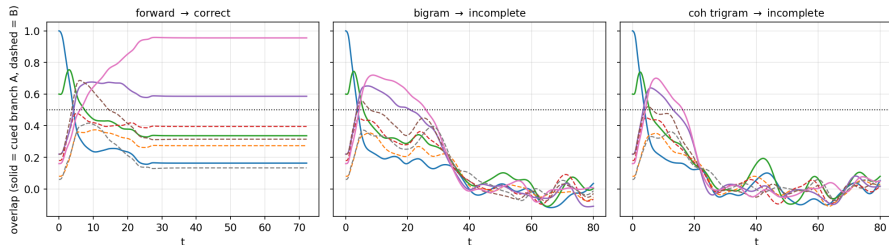
1. How much frequency? The usable band



Retrieval fraction against ω at $T = 54$ and $T = 180$, by learning rule

- $\omega = 0$ is **singular**: at an exact cue it is a fixed point, and any $\varepsilon > 0$ already sits in a different regime.
- A rotating frame **cannot** absorb a uniform ω here — the drive is anchored to the absolute $\{0, \pi\}$ lattice, so the frame shift survives in the dynamics.
- At high ω (≈ 1) the drive is overwhelmed by the natural frequency: the overlaps never peak and noise is dominant. The usable band is $\sigma_\omega \in [0.1, 0.5]$.

2. Why branching is hard, and what fixes it



Branch overlaps $m(t)$ through a 2-1-2 crossover, by learning rule

One 2-1-2 instance, cued from $A[0]$; solid = cued branch A, dashed = B; shared positions omitted.

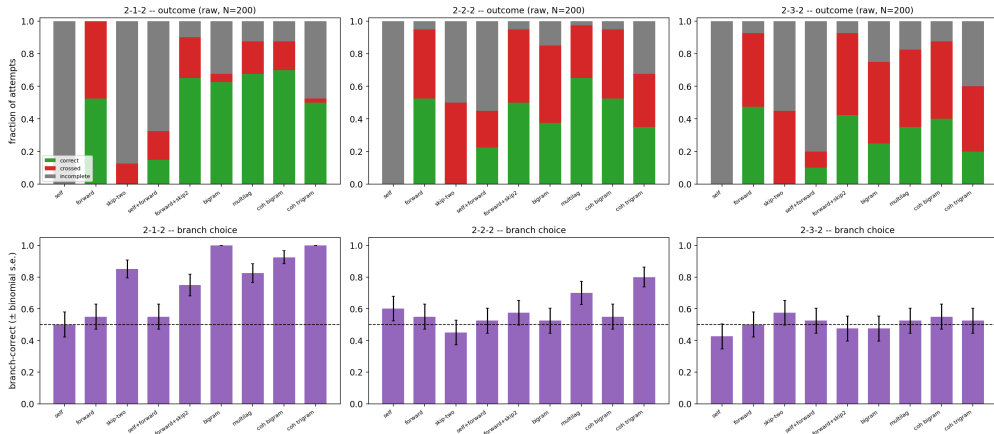
- The baseline gate reads only m_μ^d . Inside a shared run that pattern is *bit-identical* between branches, so the gate is identical and the drive points at both continuations equally — the outcome is left to residual asymmetry, i.e. chance.
- Let the gate read lags $1, 2, \dots$: if **one factor lands outside the shared run** it sees a pattern that differs between branches and the product becomes branch-specific. A run of length L needs reach $\geq L$.

3. The four crossover metrics

#	metric	definition
1	outcome	<code>correct</code> = every unshared pattern of the cued branch, <i>in order</i> , before the other appears; <code>crossed</code> = wrong branch first; <code>incomplete</code> = ran out
2	overlap quality	mean peak overlap of those patterns, over successful attempts only, with spread
3	branch-correct	did the trajectory favour the right continuation? Chance = 0.5
4	margin	signed gap between the cued and the other continuation's peak overlap

- **Metric 3 is the one that matters:** it isolates the *decision*. Metric 1 bundles it with completion, and empirically $\text{correct} \approx \text{branch-correct} \times (1 - \text{incomplete})$ to within $\sim 6\%$ — a low correct is always two failures multiplying.
- correct is also a strict AND over ~ 4 positions: 0.87 per position reads as 57%. Shared positions are excluded throughout.

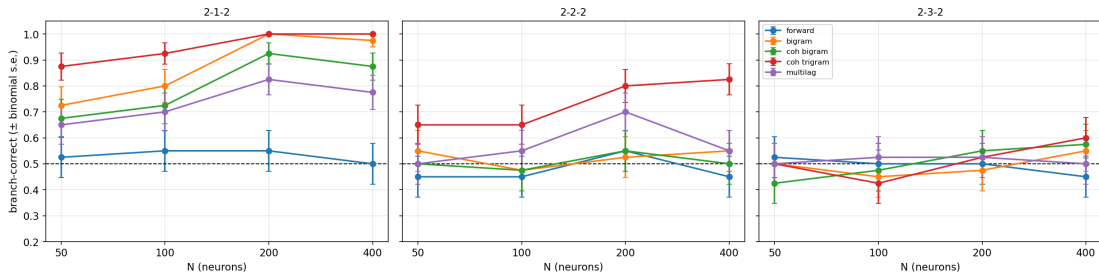
3. Benchmark across geometries



Outcome split and branch-correct rate by rule across crossover geometries

Top: outcome split. Bottom: branch choice, $\pm$ binomial standard error $\sqrt{p(1-p)/n}$. Dashed line = chance.

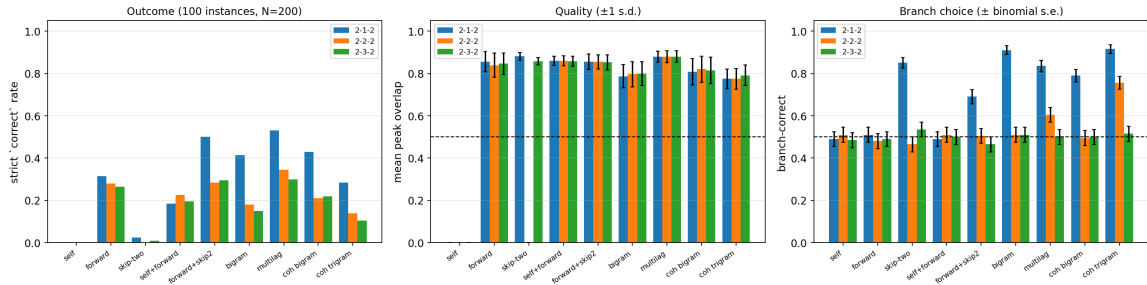
3. Branch choice improves with network size



Branch-correct rate against network size N on 2-1-2

- On 2-1-2, branch-correct rises sharply to $N = 200$: bigram $80 \rightarrow 100\%$, coh trigram $92 \rightarrow 100\%$, coh bigram $72 \rightarrow 92\%$, multilag $70 \rightarrow 82\%$. $N = 400$ is equal or slightly worse, so $N = 200$ is the operating point.
- The control works:** forward stays at chance ($55 \rightarrow 55 \rightarrow 50\%$). Network size does not manufacture branch information where the gate has none — the gain is not a decoder artifact.
- Across N , read branch-correct:** at fixed duration incomplete rises with N (a larger network needs longer to traverse), so correct understates the gain.

3. One hundred instances, with statistics



Retrieval quality and branch choice over 100 crossover instances per rule

100 crossover instances per rule, corruption rate drawn per trial from $U[0.12, 0.28]$ so each rule faces a range of difficulties; identical instances across rules. Error bars: 1 s.d. (quality), binomial s.e. (branch choice).

3. Which rule performs better

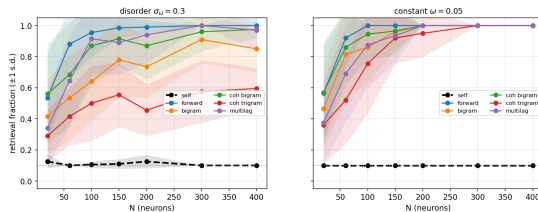
rule	reach	verdict
coh trigram	2	Only rule above chance at $L = 2$; leads at $L = 1$. Drawback is the lowest raw drive, which can be fixed with Proportional Gain.
coh bigram, bigram	1	strong at $L = 1$, fall to chance at $L = 2$ — the reach prediction, cleanly confirmed.
multilag	1 ($\times 3$)	smooth overlap evolution; drive redundancy makes it degrade gracefully.
forward	0	the control. At chance everywhere: its gate reads only the shared pattern.

- **Nothing clears chance on 2-3-2 ($L = 3$):** every rule reads at most two patterns back, so at the branch point every gate factor is still inside the shared run.
- **Reach $\geq L$ only buys a chance.** The discriminating signal is a dynamic residual surviving ~ 2 steps. Reach is cheap to build; the information is what runs out.

4. Robustness: Reading the single-chain metrics

- The crossover metrics do not carry over. On a single chain of $P = 10$: **retrieval fraction** (how much *of* a sequence came back, partial credit) vs. **strict rate** (how many *of the* sequences came back whole).
- **The $1/P$ floor.** Position 0 is the cue and is counted, so nothing scores below 10% and a rule that does nothing scores exactly that.
- **self is plotted as a control throughout.** It has no forward drive, so its curve *is* the artifact floor: any rule not clearly above it has not been shown to retrieve.
- **Robustness sweeps:** performance against network size N and corruption ratio ρ , and two knobs on the drive: proportional gain and natural frequency ω .

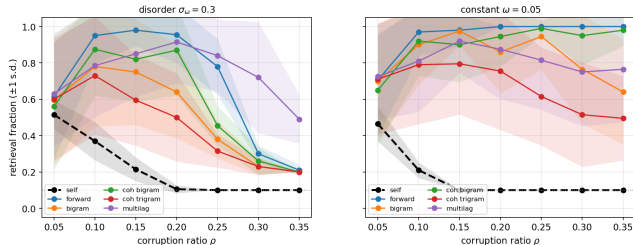
4. A retracted result: retrieval against N



Retrieval fraction against network size N at $T = 180$, heterogeneous and constant ω

- Dip in performance at $N = 200$, most likely associated with scaling of the Lyapunov landscape. This result contradicts with the exponential behavior of **Krotov's DAM** model.
- Δ retrieval fraction, $N : 100 \rightarrow 400$ — bigram $+0.21$ (disorder) / $+0.14$ (const. ω); coh bigram $+0.09$ / $+0.24$; coh bigram $+0.11$ / $+0.06$; self (control) -0.01 / $+0.00$.
- However, overall retrieval *improves* with N ; more patterns are stored without crosstalk and interference.

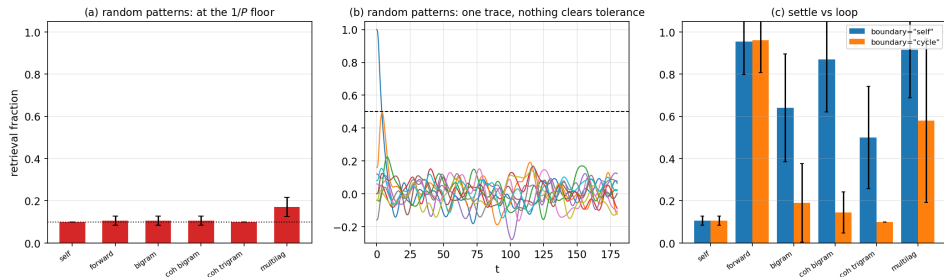
4b. U-shaped dependence on the corruption ratio



Retrieval fraction against corruption ratio ρ , heterogeneous and constant ω

- This one **holds**, in both regimes: an interior peak at $\rho \approx 0.10$ – 0.25 with failure at both ends. Small ρ : near-duplicates the decoder cannot separate. Large ρ : consecutive overlaps too small to sustain the cascade.
- **The low end inflates scores.** self peaks at the *edge*, $\rho = 0.05$, at ~ 0.5 against its 0.10 floor — pure artifact.
- **The high end is regime-dependent.** Under disorder it collapses to the floor ($\rho = 0.35$: forward 0.21, bigram 0.20); at constant ω it barely falls (forward 1.00, coh bigram 0.98). So the right-hand failure is mostly about the *spread* of ω .

5. Appendix: random patterns, and settle vs. loop



Controls: independently random patterns (a,b) and `boundary="cycle"` (c)

- **(a,b) Random patterns.** Replace the mutation chain with independently random patterns and retrieval disappears for every rule — all at the $1/P$ floor, with peak overlaps far below tolerance. Sequential similarity structure is a prerequisite.
- **(c) `boundary="cycle"` does not work.** Settling is what makes each pattern a usable attractor long enough to be decoded. Wiring the last pattern back to the first removes the terminal fixed point and the trajectory degrades rather than looping.

Findings I: learning rules

- **Reach $\geq L$ only buys a chance.** The discriminating signal is a *dynamic residual* between traversed and untraversed predecessor; it decays within ~ 2 steps. Reach is cheap to build — the information is what runs out.
- **coh trigram is the best rule by branch choice**, the only one above chance at $L = 2$; bigram/coh bigram hold at $L = 1$ and fall to chance at $L = 2$, exactly as reach predicts; multilag is the most robust.
- **Commitment against branch information** is the governing trade-off: lagged factors leave fewer for self-amplification, which is why correct and branch-correct rank the rules differently.
- **Gain correction cuts both ways:** it recovers the hardest geometry while destabilising rules that already succeeded.
- **No rule resolves $L \gtrsim 3$,** and uncorrelated random patterns give the floor for every rule.

Findings II: dynamics and scaling

- **A natural frequency is structurally necessary.** At $\omega \equiv 0$ an exact cue satisfies $\sin \theta_i = 0$: the drive vanishes identically and the system sits at an exact fixed point. Because the drive is anchored to the absolute $\{0, \pi\}$ lattice, a uniform ω is a real parameter that no rotating frame removes. The usable band $\approx [0.02, 0.2]$ is flat — more ω buys nothing.
- **Retrieval improves with N , like the dense memory it is built on,** and branch choice improves markedly with it. The earlier claim that these rules degrade with N was an *integration-time artifact* and is withdrawn.
- **U-shaped dependence on ρ :** interior peak. The high- ρ collapse is mostly the *spread* of ω and nearly vanishes at constant ω ; the low- ρ end *inflates* scores rather than depressing them.
- **A low correct is two failures multiplying,** $\text{correct} \approx \text{branch-correct} \times (1 - \text{incomplete})$. The incomplete half is a settling problem: the trajectory has parked at a sub-tolerance fixed point, and $5\times$ the time does not move it.
- **Integration time is not a free parameter.** A duration validated at one N is not validated at another — the single easiest way to manufacture a false finding here.

Tensor structure: symmetric vs. asymmetric

- [TODO: write the coupling as $J_i^{\mu_1 \dots \mu_d, \nu}$ and split it under permutation of the read indices]
- [TODO: forward rule = fully symmetric (pure-power) part; lag-context rules = mixed-symmetry part]
- [TODO: what the decomposition yields: stability conditions, or a clean statement of the reach requirement]

- [TODO: cost per step: $\mathcal{O}(KPN)$ buffer + $\mathcal{O}(\text{terms} \times N)$ drive]
- [TODO: capacity in N, d — do context terms cost capacity?]
- [TODO: Jona: scaling law / bound]

Main result

To resolve branching after a shared segment, the learning rule needs a **lag reference exceeding the length of the shared segment**. Only a factor reading back past the last shared pattern breaks the innate symmetry of the Lyapunov function; rules whose lags all sit inside the shared run are symmetric in the branches by construction.

- Markov-1 Kuramoto DAM retrieves single sequences reliably, but its Lyapunov function is symmetric at every shared state, leaving the branch choice to noise.
- Sufficient reach is only the entry ticket: commitment and branch information trade off.
- The rule alone does not fix the operating point: ρ , N and σ_ω set it *together*.

- **Physical simulators.** Every rule is pure oscillator coupling, and therefore maps onto analog and photonic Kuramoto hardware: crossover-resolving memory as a physical primitive.
- **Neurobiology.** [TODO: phase-coded replay, theta–gamma; lag-context gating as a disambiguation mechanism]
- [TODO: motor-sequence chaining, time-series disambiguation]

- ① How are these learning rules implemented in **physical (oscillator-based) simulators**?
- ② What is the role of **decaying neurons**, and can they replace the explicit lag buffer?
- ③ How does one generate **robust rules for randomly generated pattern sequences**?

Further open questions: whether the resolvable shared-run length L admits a fundamental ceiling, and whether rule weights can be *learned* rather than constructed by hand.

Question 1: physical implementation

- Rules are rank- $(d+1)$ couplings; hardware (Ising machines, OPO and polariton networks, CMOS oscillatory NNs) gives *pairwise*. Which mechanism synthesizes the product: ancilla modes, parametric mixing, or measurement feedback?
- Can the lagged factor $m_{\mu-\ell}$ be a physical *delay line* rather than a stored buffer? This would render the rule local in time as well as in space.
- **[TODO: required coupling precision and noise floor before branch choice falls to chance]**

Question 2: decaying neurons

- Replace the discrete lag by a decaying trace $\tau \dot{h}_\mu = -h_\mu + |m_\mu|^2$. Adaptation achieves the same from the opposite direction: the active pattern suppresses itself and releases the flow to its successor.
- Open: whether a single time constant τ provides reach beyond the shared segment, and whether it is simpler to realize physically than a multi-lag product.

Selected references

Delayed / slow synapses: Sompolinsky & Kanter (PRL 57, 2861, 1986); Kleinfeld (PNAS 83, 9469, 1986); Tank & Hopfield (PNAS 84, 1896, 1987). **Adaptation-driven transitions:** Horn & Usher (Phys. Rev. A 40, 1036, 1989); Herrmann, Ruppin & Usher (Biol. Cybern. 68, 455, 1993); Treves (Cogn. Neuropsych. 22, 276, 2005) — latching; Recanatesi et al. (Front. Comput. Neurosci. 9:149, 2015); Gillett, Pereira & Brunel (PNAS 117, 29948, 2020). **Closest precedent:** Martinez Mayorquin et al. (ICANN 2019, pp. 793–805) — filtered synaptic traces for *sequence disambiguation*.

Question 3: randomly generated sequences

- Uncorrelated patterns: 0% retrieval for every rule, raw and gain-corrected — peak overlaps 0.16–0.19 against a 0.5 tolerance.
- The gate fails before the drive does: for independent patterns $m_{\mu-\ell} = \mathcal{O}(N^{-1/2})$; rescaling a gate that reads $m \approx 0$ has no effect.
- To test: whitened / pseudo-inverse storage (Chaudhry et al., *Long Sequence Hopfield Memory*); larger d ; ridge-fitted weights as an upper-bound control.
- **[TODO: retrieval rate vs. pattern correlation, mutation chain → fully random: locate the collapse threshold per rule]**

Thank you
[TODO: repo QR]